Process Modeling.

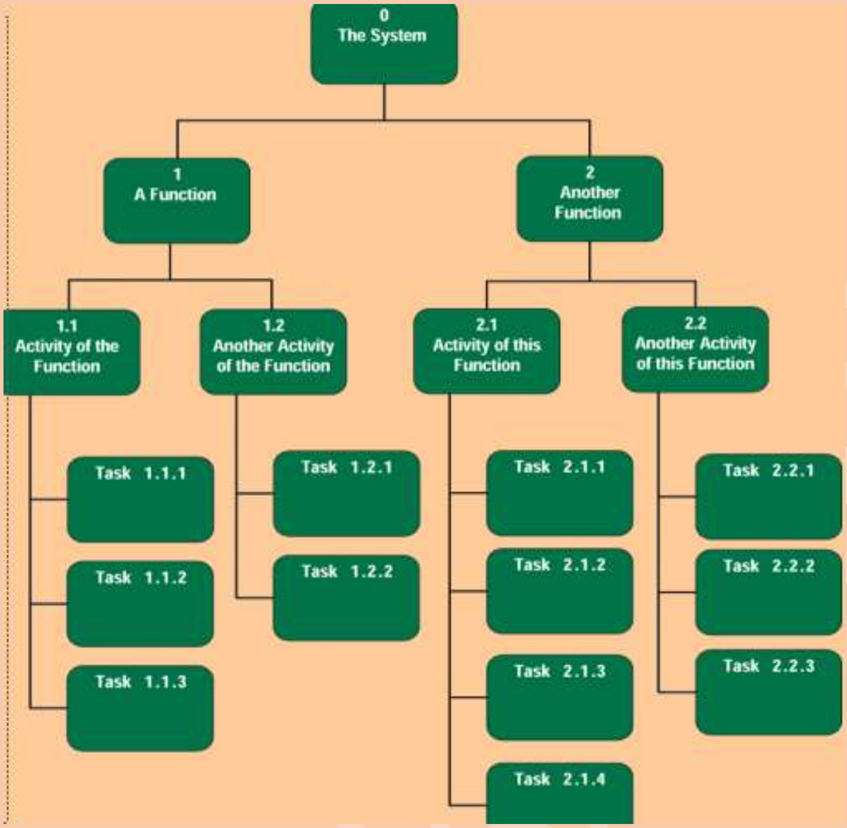
Topics:

- Model.
- Types of model.
- Process modeling.
- DFD.
    1. External agent.
    2. Data stores.
- Three symbols and one connection.
- Differences between DFDs and Flowcharts.
- Process.
- Decomposition.
- Decomposition diagram.
- Types of logical processes.
- Common process errors.
- Data flows and control.
- Data types and domains.
- Diverging and converging data flows.

- **Model:** A model is a representation of reality.
- **Types of models:**
    1. **Logical models:** Logical models show *what* a system is or does.
    2. **Physical models:** Physical models show not only what a system is or does, but also *how* the system is (to be) physically and technically implemented.
- **Process modeling:** Process modeling is a technique for organizing and documenting the structure and flow of data through a system's processes, and/or the logic, policies, and procedures to be implemented by a system's processes.
- **DFD:** A data flow diagram (DFD) is a tool (and type of process model) that depicts the flow of data through a system and the work or processing performed by that system.

1. **External agent:** An external agent defines a person, organization unit, or other organization that lies outside of the scope of the project but that interacts with the system being studied.
   External agents define the "boundary" or scope of a system being modeled
2. **Data stores:** A data store is an inventory of data.

- **Three symbols and one connection:**
  - ✓ Round rectangles represent processes or work to be done-process color
  - ✓ Square represents external agents-interface color
  - ✓ Open ended boxes represent data stores (sometimes called files or databases)
  - ✓ Arrows represent data flows, i/p and o/p to and from the process
- **Differences between DFDs and Flowcharts:**
  - Processes on DFDs can operate in parallel (at-the-same-time)
    - Processes on flowcharts execute one at a time
  - DFDs show the flow of data through a system
    - Flowcharts show the flow of control (sequence and transfer of control)
  - Processes on one DFD can have dramatically different timing
    - Processes on flowcharts are part of a single program with consistent(regular) timing.
- **Process:**A process is work performed on, or in response to, incoming data flows or conditions.
- **Decomposition:** Decomposition is the act of breaking a system into its component subsystems, processes, and subprocesses.
- **Decomposition diagram:** A decomposition diagram or hierarchy chart shows the top-down, functional decomposition of a system.

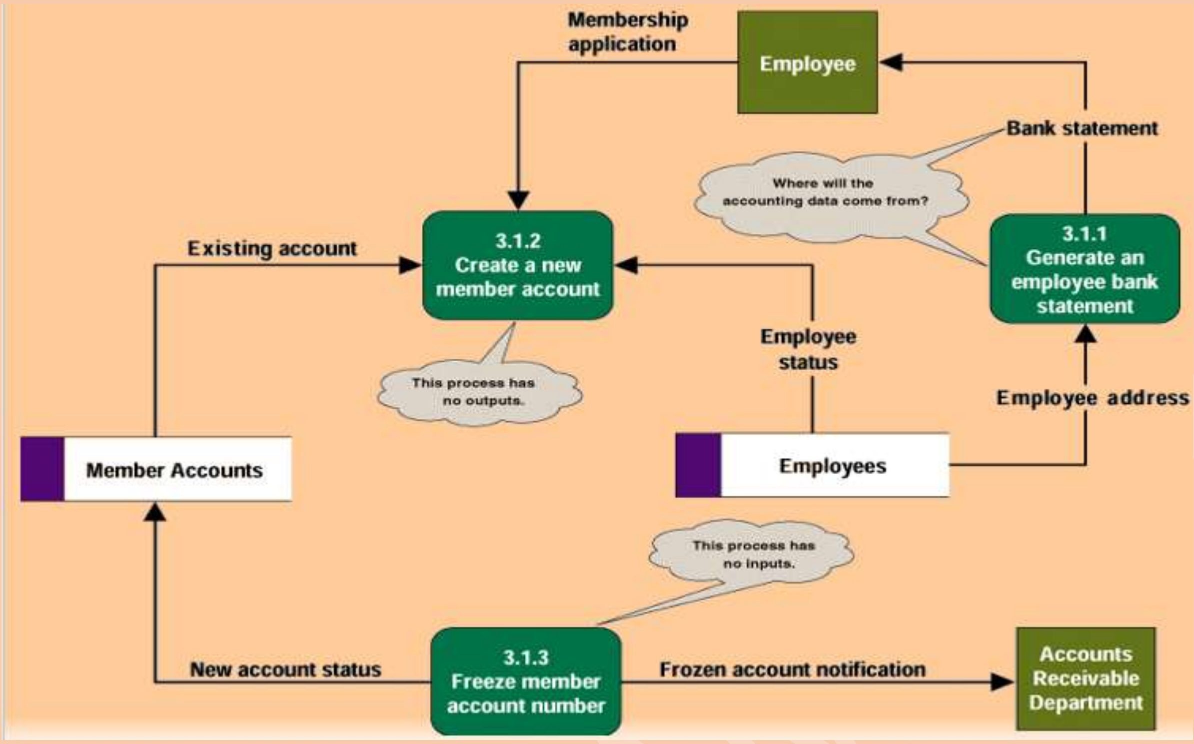


❖ **Types of logical processes:**

1. **Function:** A function is set of related and ongoing activities of a business.
2. **Event:** An event (or transaction) is a logical unit of work that must be completed as a whole (as part of a function).
3. **An elementary process:** An elementary process (or primitive process) is a discrete, detailed activity or task required to respond to an event.

❖ **Common process errors:**

1. **Black hole:** 3.1.2 has inputs but no outputs (it is called black hole because data enter the process and then disappear).
2. **David copperfield:**3.1.3 has outputs but no inputs. Unless you are David Copperfield (most commercially successful magician in history) it's a miracle.
3. **Gray hole:** 3.1.1 the inputs are insufficient to produce the o/p (it is called gray hole because
   - ✓ Misnamed process
   - ✓ Misnamed inputs and/or outputs
   - ✓ Incomplete facts



- **Data flows and control:**
  - A data flow represents an input of data to a process, or the output of data from a process.
    - A data flow may also be used to represent the creation, reading, deletion, or updating of data in a file or database (called a data store).
    - A composite data flow is a data flow that consists of other data flows.
  - A control flow represents a condition or nondata event that triggers a process.
- **Data types and domains:**
  - A data type defines what class of data can be stored in an attribute (e.g., character, integers, real numbers, dates, pictures, etc.).
  - A domain defines what values or range of values an attribute can legitimately (legally) take on.
- **Diverging and converging data flows: A diverging** data flow is one that splits into multiple data flows.

  A converging data flow is the merger of multiple data flows into a single packet.